Multistage Bilateral Noise Filtering and Edge Detection for Color Image Enhancement

Wen-Chung Kao, *Member*, IEEE and Ying-Ju Chen

Abstract —*Removing noise while preserving and enhancing edges is one of the most fundamental operations of image/video processing. When taking pictures with digital cameras, it is frequently found that the color images are corrupted by miscellaneous noise and, hence, noise filtering is necessary. The difficulty is that usually the filtering will reduce the sharpness of the image. On the other hand, optical lens imperfections are usually equivalent to spatial low pass filters and tend to result in blurred images. It is customary to apply edge enhancement algorithm on the image in order to improve the sharpness, but this process usually increase the noise level as a by-product. In this paper, we present a new integrated approach to address these issues. The proposed approach combines color edge detection, bilateral noise filter, and edge enhancement based on suitable color spaces. The experimental results show that the proposed approach can effectively reduce the noise while preserving and enhancing edges*[1].

Index Terms — Bilateral noise filtering, range filter, domain filter, edge enhancement.

I. INTRODUCTION

Additive noise removal from a given image is one of the most important image processing steps [1]-[3]. It is a practical problem faced in many digital imaging systems. Apart from the basic desire of removing noise prior to presenting the images to the observer, smoothing images also help to improve many image processing algorithms, such as compression, enhancement and recognition. An ideal noise filter aims at preserving the image details while removing the random noise. This is achieved with local adaptive recovery methods such as anisotropic diffusion, weighted least squares, and robust estimation [4], [5]. The main drawback of these methods is that they usually require several iterations to smooth out different characteristics of noise.

A good non-iterative filter called bilateral filter [6] considers both their geometric closeness and photometric similarity between the center pixel and the neighboring pixels. The bilateral filter prefers near values to distant values in both domain and range. It has been shown that bilateral filter can also be derived from some well-known noise filters using different penalty functions [7]-[10]. In spite of its achievements, bilateral filter is still an intuitive tool to get good performance on the images with obvious global edges. The global edges indicate the boundary lines between two or more large color regions but not the short ones. The bilateral filter prevents averaging the pixels across global edges, but still averages them within the same color regions of the image. Unfortunately, bilateral filter can not work well on high resolution image details such as human hairs and the fine textures of buildings in the scene. This filter highly relies on the parameter settings of the penalty functions for both domain and range filters. It is almost impossible to give a set of universal parameters or automatically determine these parameters to accommodate all scenes.

In this paper, we propose a multistage processing approach that integrates two types of noise filtering algorithms to work with two types of edge detection and one edge enhancement stage. The approach considers different characteristics of noise and edges by applying different algorithms to accommodate for respective properties. The design concepts come from the fact that the images are usually corrupted by many noise sources. It is unlikely to be able to use a simple or one stage algorithm to solve all of the problems. We carefully analyzed the noise as well as edge properties and selected suitable color space for operating the image data. The experimental results show that the proposed approach can improve the quality of the images corrupted by Gaussian-like noise.

In the following sections, we will first describe the processing flow of the proposed system. Then the noise protected edge detector with noise prefiltering is stated. In Section IV, the novel schemes to remove false edges while enhancing real edges are described. At last, the experimental results and the conclusion on the proposed noise filtering and edge enhancement approach are given in Section V and VI.

II. THE PROPOSED SYSTEM FLOW

The issues of noise filtering and edge enhancement include the following items: (a) some false edges resulted from random noise are produced by edge detectors. This is because edge points could be estimated by evaluating the maximum gradient components. Without prefiltering, many random noise may be judged as edge points by edge detectors. (b) Bilateral filter considers both geometric closeness and photometric similarity with domain and range filtering, respectively. However, it is still possible for the bilateral filter to blur some high resolution image details if a global view of edge locations is not available. (c) The edge detector for guiding bilateral noise filter provides local high frequency information, but the false edges must be identified and removed. Otherwise, the noise cannot be smoothed out in the bilateral filter stage. (d) The

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edge detector used for guiding edge enhancement should only extract longer edges but bypass the shorter ones. This is due to the fact that the human eyes are very sensitive to the enhanced edges. Shorter edges that are enhanced may look like artifacts. (e) Image noise generated from sensor or circuit imperfections are with *RGB* color space. It should be better to remove noise directly in *RGB* color space [11], [12]. Any color space transformation will change the amplitudes of the noise and result in unpredictable noise level. (f) If the three channels of color images are enhanced separately from one another, colors are corrupted in the areas close to image edges.

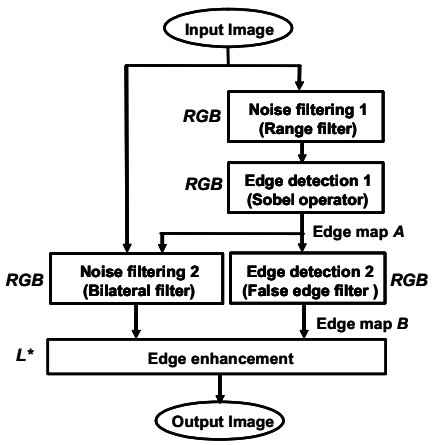

Fig. 1. The proposed noise filtering and edge Enhancement processing flow.

In order to deal with the image processing issues described above, we propose a robust processing flow to integrate edge detection, noise filtering, and edge enhancement for color images. As shown in Fig. 1, it incorporates two types of noise filter, two types of edge detector, and one edge enhancement stage. The design philosophy of the proposed approach is to solve the noise problem by using different types of noise filter or edge extractor for handling different classes of noise. The noise with amplitude not too different from that of the neighbors is filtered out in the "Noise filtering 1" stage by a range filter. The noise with amplitude much larger than that of the neighbors may not be filtered out in "Noise filtering 1" stage, but most of this type of noise will not be identified as edge points in "Edge detection 1" stage with a Sobel operator, since the Sobel operators consider the eight neighboring pixels at the same time. Unless there are two or more high amplitude noise corrupting the neighboring pixels, the Sobel operators will not identify these noise as edge points. As the amplitude of the noise has been attenuated by the range filter, only a few noise with higher amplitude can pass the screening of Sobel operators.

The edge detection result of Sobel operators is denoted as edge map *A*. Edge map *A* is not directly used for edge enhancement, but only for guiding the bilateral filter not to smooth out the edges. As described in Section I, bilateral filter performs well on large global edges of the image, but not so well for high resolution images with thin lines or textures. Given edge map *A*, the modified bilateral filter bypasses the pixels that have been identified as edge points in edge map *A* and only smoothes out the others. The edges can be kept and most of Gaussian-like noise will be removed.

Although edge map *A* is generated by two conservative stages, it is still necessary to remove some false edges before performing edge enhancement. The reason is enhancing edges of shorter length may result in some undesired artifacts. Only the edges points located on global edges are worth being enhanced. The proposed false edge filter analyzes all edge points in edge map *A* and removes all isolated edge points whose neighboring points are not edge points or only very few neighboring points are identified as edge points. The output from this false edge filter is stored in edge map *B* and only these global edge points will be enhanced.

III. Noise-Protected Edge Detector and Bilateral Noise Filter

Most of the edge detection and noise filtering algorithms are performed on the *Y* component in *YCbCr* color space or on the *L** component in *CIELAB* color space. In the proposed system, however, the noise filtering as well as edge detection algorithms are carried out in *RGB* color space. Although color image theories indicate that human eyes are more sensitive to the noise of different luminance level but not so much to chrominance component, the nature of random or fixed-pattern noise are actually produced in sensor *RGB* color space. Many color space conversion algorithms tend to increase the noise level. Hence it can be expected to obtain a better result if the noise are filtered out in an earlier stage such as the sensor *RGB* space. On the other hand, it is possible that some edges might locate at the boundary of two different chromatic blocks with similar luminance. This kind of edge might be missed if the edge detection algorithm only checks on luminance components.

As described in Section II, we combine range filter and well-known Sobel operators to determine edge map *A*. In this section, we focus on the discussion of detailed algorithms of the range filter and bilateral filter, which are directly related to the noise filtering problems.

A. The range filter

A low-pass range filter applied to an $M \times N$ image $f(x_{i,j})$ $(1 \le i \le M,\ 1 \le j \le N)$ produces an output image $h_r(x_{i,j})$ $(1 \le i \le M,\ 1 \le j \le N)$ as defined in (1), where $k_r(x)$ defined in (2) is the normalization factor for preserving the DC component after filtering, and $s(\varepsilon, x)$ defined in (3) represents the photometric similarity between the pixel at the neighborhood center x and a nearby point ε. The simplest but robust similarity function $s(\varepsilon, x)$ usually adopts shift-invariant Gaussian function of the Euclidean distance as defined in (4). The photometric spread

parameter σ_r in the image range is set to achieve the desired amount of combination of pixel values. The pixels with photometric values of distance shorter than σ_r are mixed together, but not for the pixels with distance longer than σ_r.

$$h_r(x_{i,j}) = k_r^{-1}(x) \sum_{p=i-m}^{i+m} \sum_{q=j-n}^{j+n} f(x_{i,j}) \times s(f(x_{p,q}), f(x_{i,j})) \quad (1)$$

$$k_r(x) = \sum_{p=i-m}^{i+m} \sum_{q=j-n}^{j+n} s(f(x_{p,q}), f(x_{i,j})) \quad (2)$$

$$s(\varepsilon, x) = e^{\frac{-1}{2}(\frac{\delta(f(\varepsilon), f(x))}{\sigma_r})^2} \quad (3)$$

$$\delta(f(\varepsilon), f(x)) = \left\| f(\varepsilon) - f(x) \right\| \quad (4)$$

B. The modified bilateral filter

Bilateral filter combines range and domain filter based on Gaussian kernels. The range filter has been described in Section III-A. And the domain filter considers the geometric closeness between the pixel at the center x and a nearby point ε.

A bilateral filter applied to an $M \times N$ image $f(x_{i,j})$ $(1 \le i \le M,\ 1 \le j \le N)$ produces an output image $h_b(x_{i,j})$ $(1 \le i \le M,\ 1 \le j \le N)$ as defined in (5), where $k_b(x)$ shown in (6) is the normalization factor for preserving the DC component after filtering, and $c(\varepsilon, x)$ represents the geometric distance between the pixel at the neighborhood center x and a nearby pixel ε. Similar to the range filter, the spatial distance function $c(\varepsilon, x)$ usually adopts shift-invariant Gaussian function of the Euclidean distance as shown in (7) and (8). The geometric spread parameter σ_d in the domain is chosen based on the desired amount of low pass filtering. A larger σ_d blurs the image stronger, which combines values from more distant image locations. The main trouble of the bilateral filter is that σ_d and σ_r must be adjusted according to the resolution and photometric dynamic range of the images, respectively.

$$h_b(x_{i,j}) = k_b^{-1}(x) \times \sum_{p=i-m}^{i+m} \sum_{q=j-n}^{j+n} f(x_{i,j}) \times c(x_{p,q}, x_{i,j}) \times s(f(x_{p,q}), f(x_{i,j})) \quad (5)$$

$$k_b(x) = \sum_{p=i-m}^{i+m} \sum_{q=j-n}^{j+n} s(f(x_{p,q}), f(x_{i,j})) \times c(x_{p,q}, x_{i,j}) \quad (6)$$

$$c(\varepsilon, x) = e^{\frac{-1}{2}(\frac{d(\varepsilon, x)}{\sigma_d})^2} \quad (7)$$

$$d(\varepsilon, x) = \left\| \varepsilon - x \right\| \quad (8)$$

The main difficulty with bilateral filter is that σ_d and σ_r need to be adjusted according to the resolution and photometric dynamic range of the images, respectively. To conquer the problems of the images being burred by the bilateral filter, we propose a modified version that incorporates the edge detection information from the previous stages. As shown in Fig. 1 and discussed in Section II, the edge map A shows the edge status $e_{i,j}^A$, which indicates whether the pixel $x_{i,j}$ is recognized as an edge point ($e_{i,j}^A = 1$) or not ($e_{i,j}^A = 0$). By incorporating this edge information, the output $h(x_{i,j})$ of the modified bilateral is defined in (9).

$$h(x_{i,j}) = \begin{cases} h_b(x_{i,j}) & if \quad e_{i,j}^A = 1 \\ f(x_{i,j}) & if \quad e_{i,j}^A = 0 \end{cases} \quad (9)$$

IV. Edge Enhancement

Although most of edges can be preserved by the modified bilateral filter, the optical lens imperfections will cause the image to be blurred, especially at the four corners. Hence it is necessary to enhance the edges in the image processing pipeline.

The main issues of edge enhancement include the following items: (a) The boundary effects of enhancing smaller edges are unpredictable because most of edge detectors as well as noise filters work on the local image data. They lack a full view of the distribution of longer edges. (b) If the tri-stimulus values *RGB* of the image pixel are enhanced separately, it is possible to introduce unexpected colors or artifacts.

In the following subsections, we will present the approaches to solve the issues listed above. The improvements are due to the fact that we use a strict strategy to identify the edges before enhancing them. In addition, the edge enhancement is performed on the edge pixels under *CIELAB* color space, which is a uniform color space with luminance and chrominance components separated. Only the luminance components (L^*) of the edge pixels are enhanced, and the chrominance values (a^* and b^*) of the edge pixels are smoothed out. It prevents the edge enhancement from introducing color artifacts.

A. False edge filter

Although range filter can smooth out the image before applying Sobel operators, some false edges may also be extracted. The Sobel operators perform well in extracting all possible edges in the image, but the noise with higher amplitude will also be identified as edges. The proposed false edge filter analyzes the density $\omega_{i,j}$ of the edge pixels

surrounding the current point $x_{i,j}$. Only the edge pixels that locate in the cluster of edge pixels in edge map A are selected ($e^B_{i,j}=1$) for enhancement in the next stage. Any isolated edge pixels in edge map A will be recognized as normal pixels ($e^B_{i,j}=0$) in edge map B. The above concept can be formulated as (10) and (11), where a and b is the window size for counting edge pixels, and μ is the predefined parameter for controlling the strength of the false edge filter.

$$e^B_{i,j}=\begin{cases}1 & if\ \omega_{i,j}\geq\mu\\ 0 & if\ \omega_{i,j}<\mu\end{cases} \tag{10}$$

$$\omega_{i,j}=\sum_{i-a}^{i+a}\sum_{j-b}^{j+b}e^A_{p,q} \tag{11}$$

B. Edge enhancement

Most of edge enhancement algorithm works on standard *RGB* (*sRGB*) or *YCbCr* color space. The luminance and chrominance components are not separated exactly with these color spaces. Changing any one of the components in *YCbCr* will change both chrominance and luminance simultaneously. With *sRGB* or *YCbCr* color spaces, it is almost impossible to keep predicable colors through edge enhancement.

The *CIELAB* color space is based on a color vision model. This space is transformed from *CIEXYZ* tri-stimulus values into an achromatic lightness values L^* and two chromatic values a^* and b^* using the transformations shown in (12) – (15), where X_n, Y_n, and Z_n are the tri-stimulus values of the reference (neutral) white point, and $g(t)$ is defined in (16).

$$\begin{bmatrix}X\\Y\\Z\end{bmatrix}=\begin{bmatrix}0.412453 & 0.357580 & 0.180423\\ 0.212671 & 0.715160 & 0.072169\\ 0.019334 & 0.119193 & 0.950227\end{bmatrix}\begin{bmatrix}R_s\\G_s\\B_s\end{bmatrix} \tag{12}$$

$$L^*=116\ \ g\left(\frac{Y}{Y_n}\right)-16 \tag{13}$$

$$a^*=500\left(g\left(\frac{X}{X_n}\right)-g\left(\frac{Y}{Y_n}\right)\right) \tag{14}$$

$$b^*=200\left(g\left(\frac{Y}{Y_n}\right)-g\left(\frac{Z}{Z_n}\right)\right) \tag{15}$$

$$g(t)=\begin{cases}t^{1/3} & t>0.008856\\ 7.787t+\dfrac{16}{116} & t\leq 0.008856\end{cases} \tag{16}$$

The inherent properties of *CIELAB* color space are that radial distance and angular position represent the chroma and hue of the color, respectively. It is much easier to handle color consistency and brightness in *CIELAB* color space than in other types of color space such as *sRGB* or *YCbCr*. For each edge pixels, the gradient of luminance values on edge pixels can be enlarged by changing the L^* component only, whereas hue and color saturation values are kept unchanged.

As shown in Fig. 2, the concept of edge enhancement (also called sharpening) is to highlight fine details by enlarging the luminance difference between the two sides of the boundary corresponding to the processed edge. The sharpening concepts described above can be realized by (17) and (18), where $L(x_{i,j})$ and $L^E(x_{i,j})$ represent the original and the enhanced stimulus value L^* of the pixel $x_{i,j}$, respectively. The edge status $e^B_{i,j}$ indicates whether the pixel $x_{i,j}$ is recognized as an edge point ($e^B_{i,j}=1$) or not ($e^B_{i,j}=0$) in edge map B. The enhanced values can be determined by the percentage ϕ of its luminance value.

$$L^E(x_{i,j})=\begin{cases}L(x_{i,j})+\phi L(x_{i,j}) & if\ \ (e^B_{i,j}=1)\ \ and\ \ (L(x_{i,j})>\theta_{i,j})\\ L(x_{i,j})-\phi L(x_{i,j}) & if\ \ (e^B_{i,j}=1)\ \ and\ \ (L(x_{i,j})<\theta_{i,j})\\ L(x_{i,j}) & if\ \ (e^B_{i,j}=0)\ \ or\ \ (L(x_{i,j})=\theta_{i,j})\end{cases} \tag{17}$$

$$\theta_{i,j}=\frac{1}{9}\sum_{i-1}^{i+1}\sum_{j-1}^{j+1}L(x_{p,q}) \tag{18}$$

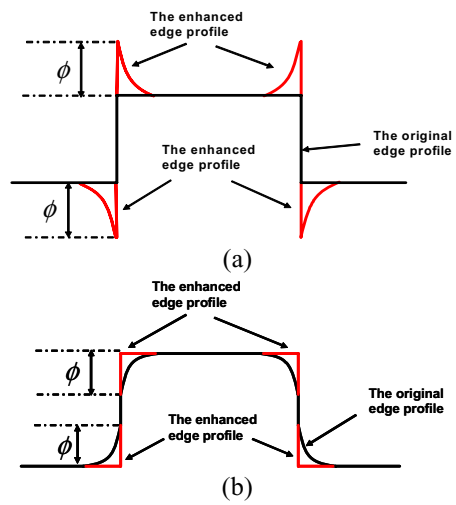

Fig. 2. The edge enhancement for (a) the non-blurred edge and (b) the blurred edge.

Note that the edge points may be corrupted by random noise, but they are not smoothed out in the "Noise filter 2" stage as formulated in (5)-(9). If the luminance components L of the edge points are enhanced, the noise may also be enlarged. To reduce these undesired color artifacts, the chrominance components a^* and b^* of the edge point $x_{i,j}$ must be smoothed out to their new values $a^E(x_{i,j})$ and $b^E(x_{i,j})$ when the edge point is enhanced. The concept can be realized by (19) and (20), where $a(x_{i,j})$ and $b(x_{i,j})$ represent the original stimulus value a^* and b^* of the pixel $x_{i,j}$, respectively.

$$a^E(x_{i,j})=\frac{1}{9}\sum_{p=i-1}^{i+1}\sum_{q=j-1}^{j+1}a(x_{i,j}) \tag{19}$$

$$b^E(x_{i,j})=\frac{1}{9}\sum_{p=i-1}^{i+1}\sum_{q=j-1}^{j+1}b(x_{i,j}) \tag{20}$$

V. EXPERIMENTAL RESULTS

Fig. 3 The filtering result by the bilateral filter. (a) (σ_d, σ_r)=(1,10), (b) (σ_d, σ_r)=(1,90), (c) (σ_d, σ_r)=(10,90), and (d) (σ_d, σ_r)=(10,90).

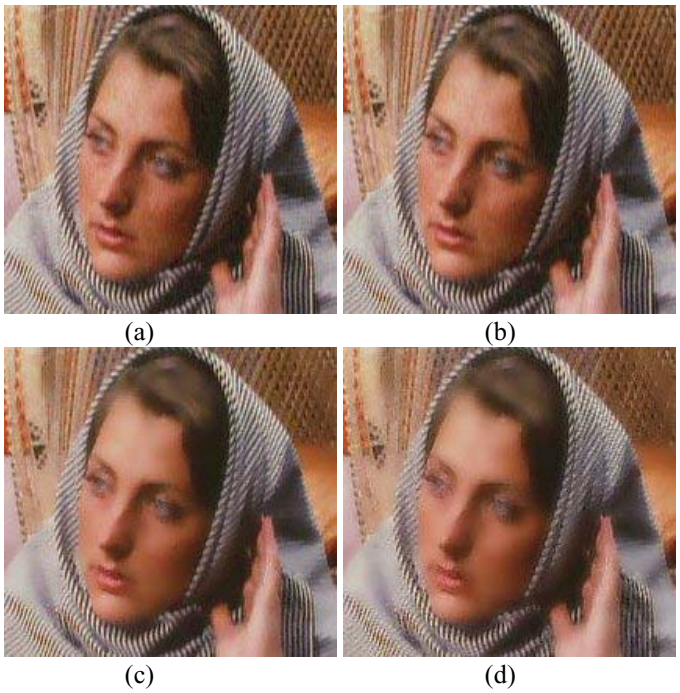

Fig. 4. The filtering result by the proposed system. (a) (σ_d, σ_r)=(1,10), (b) (σ_d, σ_r)=(1,90), (c) (σ_d, σ_r)=(10,10), and (d) (σ_d, σ_r)=(10,90).

Accurate detection of edges in noisy data needs to meet two possibly conflicting requirements. The edge detection process should avoid false edge produced by noise while ensuring correct edge detection. In order to demonstrate the performance of the proposed algorithms of noise filtering and edge enhancement, we have performed computer simulations to deal with many noise-corrupted images. The noisy images are generated by adding zero-mean Gaussian noise with standard deviation $\sigma = 15$.

Fig. 3 and Fig. 4 show the results that are processed by bilateral filter only and our approach, respectively. In these tests, we set the several parameters for both the photometric spread parameter σ_r and geometric spread parameter σ_d used in bilateral filter and our system. All of the filtering window used in the range filter, bilateral filter and false edge filter are set to 9×9 for these two approaches. While comparing the results, we can easily find the proposed approach shows better result on the thin lines or textures. The bilateral filter shows good performance on noise filtering, but many image details are also removed. By using the proposed multistage strategies to identify the edges and smooth out the noise, the image lines can be kept, but most of the image noise can be filtered out.

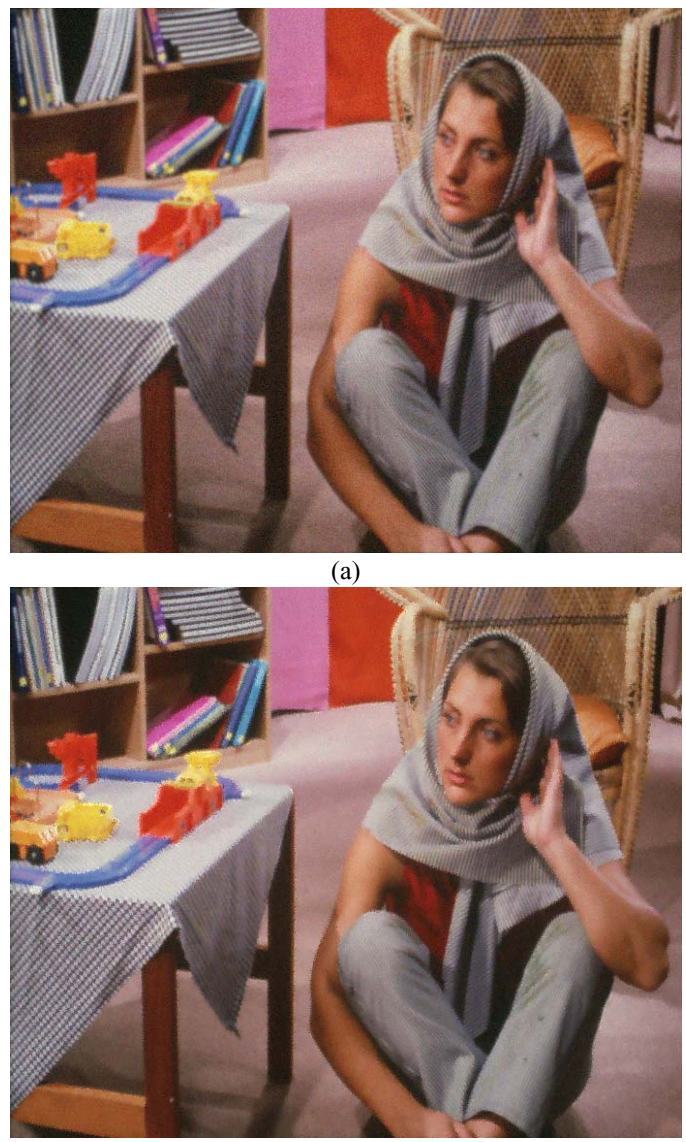

Fig. 5. Edge enhancement result: (a) The defocused and noise corrupted image, (b) (σ_d, σ_r, ϕ)=(5,10,0.05).

On the other hand, due to the imperfections of the optical lens, the image may be blurred. The effect of which is similar to a spatial low pass filter. For demonstrating the reconstruction capability of the proposed system, an example of defocused with noise corrupted image and the processing result from our system are shown in Fig. 5. The tested noisy image is generated by using a 3×3 filtering window to simulate the effect of optical lens defocused, and then adding zero-mean Gaussian noise with standard deviation $\sigma = 15$. Note that human eyes are very sensitive to the artifacts. Introducing any false colors will be easily found in the pictures. Since our system verifies the extracted edges conservatively, only the longer edges are enhanced. The enhancement process is done only on the L^* components in *CIELAB* color space. Hence, the edges are enhanced in the blurred image and artifacts are minimized.

VI. Conclusion and Discussion

Removing image random noise is one of the most crucial one for designing digital still cameras. It is particular true for the cases that the analog signal gain is set as a higher value when taking pictures in extreme dark scenes. Although it can remedy the problem happening in such scenes by taking pictures with longer exposure time, the possible vibrations by hands will result in image burred.

In this paper, multistage noise filtering with edge enhancement approach, which integrates bilateral filtering and several edge detection algorithms for extraction of edges and filtering out the noise, has been presented. The main improvements over traditional approach include: (1) several noise filters are adopted to deal with different classes of noise and (2) different color spaces are used to handle the problems of noise filtering and edge enhancement. Although the computation complexity of the proposed approach is higher than traditional bilateral filter, the design concept can be realized by hardware design which will be our future work.

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